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CSC360 - Computer Graphics and Digital Image Processing

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Reflection: Computer Graphics & Digital Image Processing

Today's class covered a dual-focused overview that connected essential system operations and network security with the fundamental concepts of digital media processing.

Part 1: System, Security & Networking Basics

This part involved tackling the basics of secure remote communications. SSH (Secure Shell) and its security compared to HTTP were studied in detail. While HTTP uses cleartext communication that is susceptible to any interception until it is encrypted using HTTPS, SSH is end-to-end encryption, authentication and secure data transfer over a network.

- Public Key: Known to everyone used for message encryption or signature verification.
- Private Key: Only known to its holder used for decryption or signing messages.

The practical use of this knowledge involved learning about how to set up an authenticated environment including how to create the .ssh directory for storage of keys and config files. Basic directory navigation was also covered, where the home directory was described as the main user workspace in an operating system.

Part 2: Foundations of Digital Media & Image Processing

The second part of the lesson was dedicated to the key topics of image processing and graphics:

- Pixel (Picture Element): The smallest programmable unit of a digital screen, containing color and intensity values.
- Digital Image: A two-dimensional array of pixels stored in numeric form (generally in matrix form as color channel values such as RGB).
- Animation: The sequential display of static 2D or 3D images at rapid succession to create an optical illusion of motion.
- Video File: A file that contains a sequence of digital images (frames) in sync with audio signals and other meta-data compressed using various algorithms.
- Interpolation: A mathematical process of estimating values between known values—it is essential in graphics for the purposes of resizing images and converting frames per second of videos.

Key Point

In order to be able to effectively manage processing pipelines, remote servers, and the storage of data sets within current graphics and image processing technologies, it is a prerequisite to secure workflows using methods such as SSH, as well as understand basic directory structures. It is essential to establish a solid foundation regarding pixels and mathematical operations such as interpolation.